

Research on Load Forecasting and Dispatching Strategy of Virtual Power Plant Based on Big Data

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Abstract—This study explores a big data-driven approach for load forecasting and scheduling in virtual power plants (VPPs), proposing a dual-layer framework that integrates predictive analytics with dynamic scheduling optimization. In the forecasting stage, a hybrid model combining LSTM and XGBoost is employed for short-term load prediction. By leveraging LSTM's strength in capturing time series patterns and XGBoost's ability to extract nonlinear features, the model achieves notable improvements in prediction precision. In the scheduling stage, dynamic optimization strategies are developed using mixed integer linear programming (MILP) alongside deep reinforcement learning (DRL), aiming to cut operational costs and enhance the integration of renewable resources. Simulation results confirm the method's effectiveness: the LSTM-XGBoost model reaches a mean absolute percentage error (MAPE) of 5.1% and a root mean square error (RMSE) of 312.8 kW, outperforming traditional models in accuracy. Moreover, the MILP-DRL-based scheduling approach yields cost reductions of 7.5% under clear skies and 12.0% in variable weather conditions, while significantly boosting solar and wind energy utilization and lowering overall curtailment rates.

Keywords—big data, load forecasting, scheduling strategy, virtual power plant, LSTM-XGBoost, mixed integer linear programming, deep reinforcement learning

I. INTRODUCTION

Virtual power plant (VPP) improves the flexibility and stability of power system by integrating distributed energy resources, examples include solar generation, wind energy systems, energy storage units, and flexible loads. These technologies offer effective solutions to address mismatches between energy production and consumption, while also supporting greater integration of renewable resources. In virtual power plant (VPP) operations, forecasting demand and formulating scheduling plans are crucial. Precise demand predictions provide a sound basis for grid dispatch, enabling better resource optimization and cost reduction. Meanwhile, well-designed scheduling ensures the power system operates safely and steadily, enhancing supply reliability. However, the traditional load forecasting and scheduling methods are often difficult to adapt to the complex and changeable operating environment of VPP, and the forecasting accuracy and scheduling efficiency need to be improved.

Power load forecasting uses historical data to build a forecasting model to improve the forecasting accuracy. Power equipment scheduling improves scheduling efficiency

and avoids economic losses through intelligent means [1-2]. Power system optimal dispatching integrates advanced information and communication technology to improve energy utilization efficiency [3]. Neural network method uses neural network to construct prediction model, which has adaptive ability and intelligent processing ability [4]. The Support Vector Machine (SVM) technique is grounded in statistical learning theory and is known for its robust generalization capabilities [5]. Meanwhile, regression analysis explores the relationship between independent and dependent variables, enabling predictions about future load values [6]. Fuzzy forecasting method combines fuzzy mathematics theory to deal with the uncertainty of load change [7]. The study of VPP load forecasting and dispatch strategies utilizing big data is crucial for optimizing the performance of power systems. As technologies like the Internet of Things, big data, and artificial intelligence continue to evolve, the integration and smart capabilities of VPPs are expected to improve significantly [8]. VPP will participate more actively in electricity market transactions, and maximize its value by providing auxiliary services such as frequency modulation and peak shaving [9].

Big data technology can uncover the underlying patterns in power load variations, providing more precise data for load forecasting. Additionally, it enables real-time monitoring and intelligent scheduling of power systems, enhancing the relevance and effectiveness of scheduling strategies. Consequently, this paper explores the load forecasting and scheduling approach for Virtual Power Plants (VPPs) based on big data, aiming to boost forecasting accuracy and scheduling efficiency while offering technical support for the optimal operation of VPPs.

II. METHODOLOGICAL FRAMEWORK

This study adopts a two-tier architecture of data-driven prediction and dynamic optimal scheduling, as shown in Figure 1. The data layer integrates multi-source heterogeneous data inside and outside VPP. The forecasting stage utilizes a combination of the LSTM and XGBoost models for short-term load prediction [10-11]. Based on these predictions, the scheduling phase employs a mix of mixed integer linear programming (MILP) and deep reinforcement learning (DRL) techniques to formulate a dynamic scheduling approach.

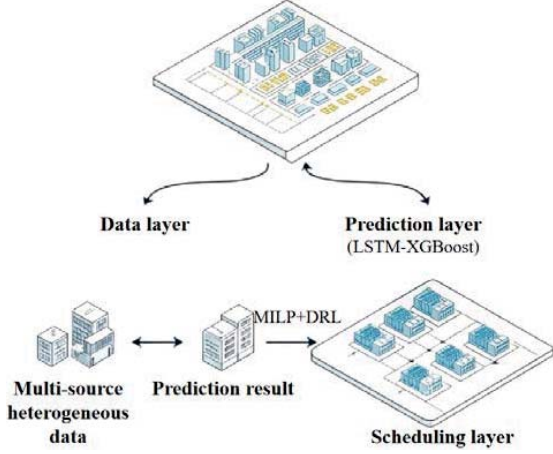


Fig. 1. Two-tier architecture of data-driven prediction and dynamic optimal scheduling

III. CONSTRUCTION OF VPP LOAD FORECASTING MODEL BASED ON BIG DATA

Traditional load forecasting methods mainly rely on statistical analysis and time series analysis, and these methods perform well in dealing with simple and stable load changes. However, with the continuous expansion of power system scale and the wide access of distributed energy resources, the load change presents more complex and changeable characteristics. In the face of these complex changes, it is often difficult for traditional methods to accurately capture the inherent law and changing trend of load, which leads to the decline of forecasting accuracy.

Traditional load forecasting methods usually rely on manual experience and expert knowledge, and the utilization of data is limited. In the age of big data, the availability of vast amounts of power-related data offers a richer set of information for forecasting energy demand, but it is difficult for traditional methods to effectively mine and utilize valuable information in these data, thus limiting the improvement of forecasting ability.

To address the shortcomings of traditional load forecasting techniques, this paper introduces a VPP load forecasting framework that leverages big data. This approach fully utilizes big data technologies for efficient data processing, feature extraction, and model training, and reveals the inherent law of load change by mining and analyzing massive power data, thus improving the forecasting accuracy.

The load forecasting model adopts LSTM-XGBoost fusion architecture, which combines the time series modeling ability of LSTM and the nonlinear feature mining ability of XGBoost (Figure 2). LSTM dynamically controls information flow through its gating mechanism, and captures long-term dependence in time series by using cell state and hidden layer state. Forgetting gate decides how much past information to keep, input gate regulates the writing of new information, output gate generates the output at the current moment, and nonlinear transformation and normalization are realized with the help of Sigmoid and Tanh activation function.

XGBoost optimizes the objective function to improve the prediction accuracy, and its objective function consists of

prediction error loss and regularization term. The prediction error loss quantifies the difference between the actual and predicted values, while the regularization term helps prevent overfitting by controlling the complexity of the model. The objective function for XGBoost can be expressed as follows:

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (1)$$

Where $l(y_i, \hat{y}_i)$ is the loss of prediction error. $\Omega(f_k)$ is a regularization term, which controls the complexity of the tree. Among them,

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (2)$$

Where T is the number of leaf nodes of the tree and w is the leaf weight.

IV. DESIGN OF VPP SCHEDULING STRATEGY BASED ON BIG DATA

A. Economic dispatch objective function

Power system dispatching is a crucial process that ensures the equilibrium between electricity supply and demand, while also maintaining the system's stability. It involves coordinating and optimizing various aspects such as power generation, transmission, and distribution. This is done by considering the real-time operating conditions and forecasting information, with the goal of fulfilling user demand for electricity while ensuring the system remains efficient and reliable.

In power system dispatching, it is necessary to follow the constraints of supply and demand balance to ensure the matching of power generation and electricity consumption and ensure the adequacy of power supply; It is also necessary to consider the unit start-stop constraints to ensure that the operation is in accordance with technical and economic laws and maintain the stability and reliability of the system. Transmission network constraints require that the transmission capacity and stability of power lines must be considered in scheduling to prevent overload and failure. With the enhancement of environmental awareness, the dispatching strategy should also take into account environmental protection requirements such as reducing pollution emissions and improving the utilization rate of renewable energy.

The objective function of economic dispatch is realized by MILP, which aims at minimizing the total operating cost, including the unit generation cost, power grid purchase cost and power deviation penalty term [12]. This method comprehensively considers the time-of-use price and forecasting error, and optimizes the unit output and power purchase plan in each period.

$$\min \sum_{t=1}^T (C_i^{gen} P_{i,t} + C_t^{grid} P_t^{grid} + C^{penalty} \cdot \Delta P_t) \quad (3)$$

Where C_i^{gen} is the power generation cost of the i unit.

$P_{i,t}$ is the output of unit i in time period t . C_t^{grid} is the power grid purchase cost. ΔP_t is the penalty term of power deviation.

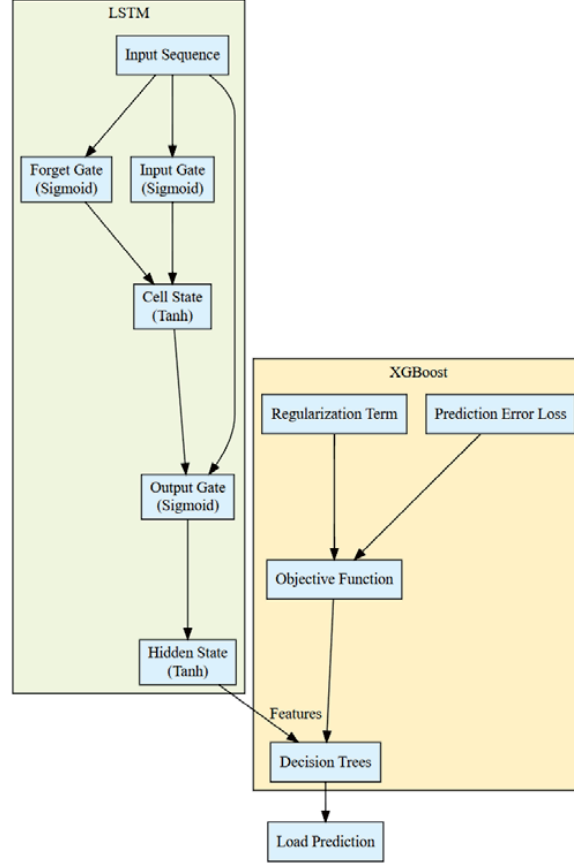


Fig. 2. LSTM-XGBoost fusion architecture

B. Scheduling strategy of deep reinforcement learning

The VPP scheduling strategy, powered by big data, enhances the management of distributed energy resources by providing precise load forecasts. This leads to improved system efficiency and reliability, as well as cost-effectiveness. The future power demand is predicted by using the load forecasting model, and the dispatching plan matching with the actual demand is made according to this. Under the premise of meeting the constraints of unit start-stop, transmission network and environmental protection, the power generation arrangement of each distributed energy source is planned, the power generation plan is continuously adjusted in real-time based on monitoring to maintain system stability and responsiveness. This approach not only accommodates fluctuations in demand but also maximizes the use of renewable energy, helping to decrease environmental pollution.

In order to deal with the complex and changeable power system scheduling problem, this paper introduces DRL to realize the optimal control of VPP. The DRL scheduling strategy is based on the Q-learning framework, and the scheduling decision is optimized by constantly updating the action value function (Q value) [13-14]. The state space integrates information such as load forecasting, renewable energy output and energy storage state, and the action space covers unit start-stop and energy storage charge-discharge

instructions. The model uses discount factor and learning rate to balance long-term income and short-term adjustment, and realizes efficient adaptive scheduling in complex and changeable operating environment, which improves the flexibility and reliability of the whole system.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \quad (4)$$

Where s_t is the state space, such as load forecast value, renewable energy output and energy storage SOC. a_t is the action space, including unit start-stop and energy storage charge-discharge instructions. γ is the discount factor and α is the learning rate.

V. SIMULATION EXPERIMENT AND RESULT ANALYSIS

A. Simulation platform

In software, Python (PyTorch, XGBoost library) is used to realize the LSTM-XGBoost hybrid forecasting model, and MATLAB/Simulink is used to build the VPP with photovoltaic, wind power, energy storage and adjustable load. Gurobi is used to solve the MILP problem and TensorFlow is

used to realize the DRL scheduling strategy. In terms of hardware, MySQL database is used to store historical load, weather and equipment operation data, and Alibaba Cloud ECS server (4-core CPU+16GB memory) is used to realize minute-level real-time calculation and dispatch instructions, which provides efficient support for system simulation.

B. Experimental design

Focusing on the operation data of a coastal VPP in summer of 2024, the peak load reaches 50MW, of which 30% is photovoltaic and 20% is wind power. In order to evaluate the effectiveness of the proposed LSTM-XGBoost hybrid forecasting model and MILP-DRL hybrid scheduling strategy, a comparative experiment was set up. In the aspect of prediction, the traditional ARIMA model and BP neural network (BPNN) are compared with the mixed model in this paper. On the scheduling strategy, the effects of traditional MILP method and MILP-DRL hybrid strategy are compared.

For the forecasting model, the input characteristics include historical load, temperature, humidity, wind speed and electricity price, etc. A 24-hour time window is adopted, and the training set and test set are divided according to the ratio of 70%/30%. At the same time, the sliding window cross-validation is applied to enhance the generalization ability of the model. According to the dispatching model, the weight of objective function is set, that is, the cost of power generation accounts for 0.6, the cost of power purchase accounts for 0.3, the penalty term accounts for 0.1, and the dimension of DRL state space is set to 12 dimensions, including key factors such as SOC of energy storage, load

deviation forecast and wind and solar output fluctuation.

The experimental implementation is divided into two typical scenarios: scheduling experiments for three consecutive days in sunny days, when the photovoltaic output is relatively stable; As well as the dispatching experiment for three consecutive days under rainy and windy weather conditions, the wind and scenery output fluctuates greatly in this scene.

C. Results analysis and discussion

The comparison of load forecasting performance (as shown in Table I) shows that the LSTM-XGBoost hybrid model is the best in all indicators, with an average absolute percentage error (MAPE) of 5.1% and a root mean square error (RMSE) of 312.8 kW. Although the training time takes 40 minutes, it is compared with the 8.2% MAPE and 385.2 kW RMSE of ARIMA model and 6.7% of BPNN. This shows that although the training of LSTM-XGBoost model takes a long time, it has obvious advantages in improving the prediction accuracy. LSTM captures the time series dependence, and XGBoost enhances the nonlinear fitting ability of features (Figure 3).

TABLE I. COMPARISON OF LOAD FORECASTING PERFORMANCE

Model	MAPE (%)	RMSE (kW)	Training time (min)
ARIMA	8.2	420.5	5
BPNN	6.7	385.2	25
LSTM-XGBoost	5.1	312.8	40

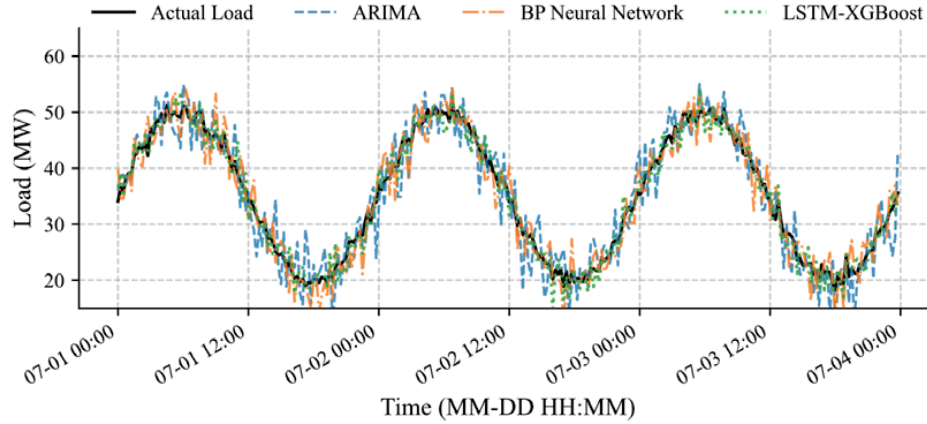


Fig. 3. Comparison of load forecasting curves

The economic comparison of scheduling strategies shows that the cost of MILP-DRL hybrid scheduling is 172,000 yuan in sunny days, which is 7.5% lower than the traditional MILP scheduling of 186,000 yuan. In the scenario with large weather fluctuation, the scheduling cost of MILP-DRL is

214,000 yuan, which is 12.0% lower than that of the traditional method of 243,000 yuan. This shows that MILP-DRL scheduling strategy can achieve significant cost savings in both stable and fluctuating environments. See Table II for the economic comparison of dispatching strategies.

TABLE II. ECONOMIC COMPARISON OF DISPATCHING STRATEGIES

Scene	Traditional MILP scheduling cost (ten thousand yuan)	MILP-DRL scheduling cost (ten thousand yuan)	Cost reduction
Sunny day (scene 1)	18.6	17.2	7.5%
Fluctuation (Scenario 2)	24.3	21.4	12.0%

Table III shows that MILP-DRL scheduling strategy can significantly improve the utilization rate of photovoltaic and wind power, while reducing the total energy rejection rate. Under MILP-DRL strategy, the photovoltaic utilization rate reached 91.3%, and the wind power utilization rate reached

87.6%. In contrast, the photovoltaic and wind power utilization rates of traditional MILP strategy were 82.1% and 78.5% respectively. The total energy rejection rate of MILP-DRL strategy is only 3.2%, which is far lower than the traditional method's 9.4%, which shows the superiority of

this strategy in improving the utilization efficiency of renewable energy. Fig. 4 shows that the SOC variation curve

has a higher matching degree with the wind and light output.

TABLE III. ANALYSIS ON THE ABSORPTIVE CAPACITY OF RENEWABLE ENERGY

Tactics	Photovoltaic utilization rate (%)	Utilization ratio of wind power (%)	Total energy rejection rate (%)
Traditional MILP	82.1	78.5	9.4
MILP-DRL	91.3	87.6	3.2

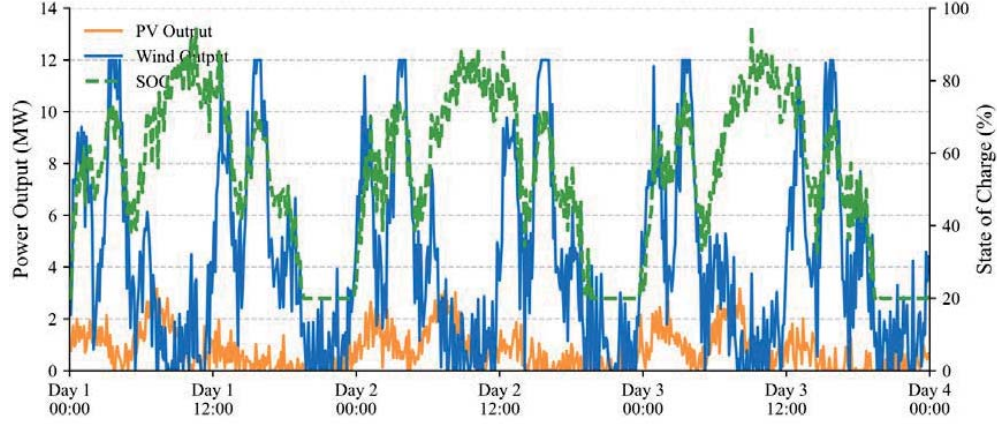


Fig. 4. Energy storage SOC and wind and light output curve

VI. CONCLUSION

A dual-layer framework is established for data-driven forecasting and dynamic optimization of scheduling. The short-term load prediction is carried out using the LSTM-XGBoost hybrid model, while a dynamic scheduling approach is formulated by integrating MILP and DRL techniques. The experimental findings demonstrate that the LSTM-XGBoost hybrid model outperforms the traditional ARIMA model and BPNN in terms of prediction accuracy. Although the training time is long, it provides higher prediction accuracy. In terms of scheduling strategy, MILP-DRL hybrid strategy is excellent in cost saving and renewable energy utilization, and can achieve significant cost reduction and energy abandonment rate reduction in both stable and fluctuating environments. This approach greatly enhances the efficiency of utilizing photovoltaic and wind power, while simultaneously decreasing the overall rate of energy waste. It demonstrates its advantage in boosting the effective use of renewable energy sources. The big data-based load forecasting and scheduling method for Virtual Power Plants (VPP) introduced in this research significantly enhances forecasting accuracy and scheduling effectiveness, offering robust technical support for the optimal functioning of VPPs.

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